

ECE 383

FINAL PROJECT

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JETPACK JOYRIDE + NES



WHY?

LIFE WITH JETPACK JOYRIDE



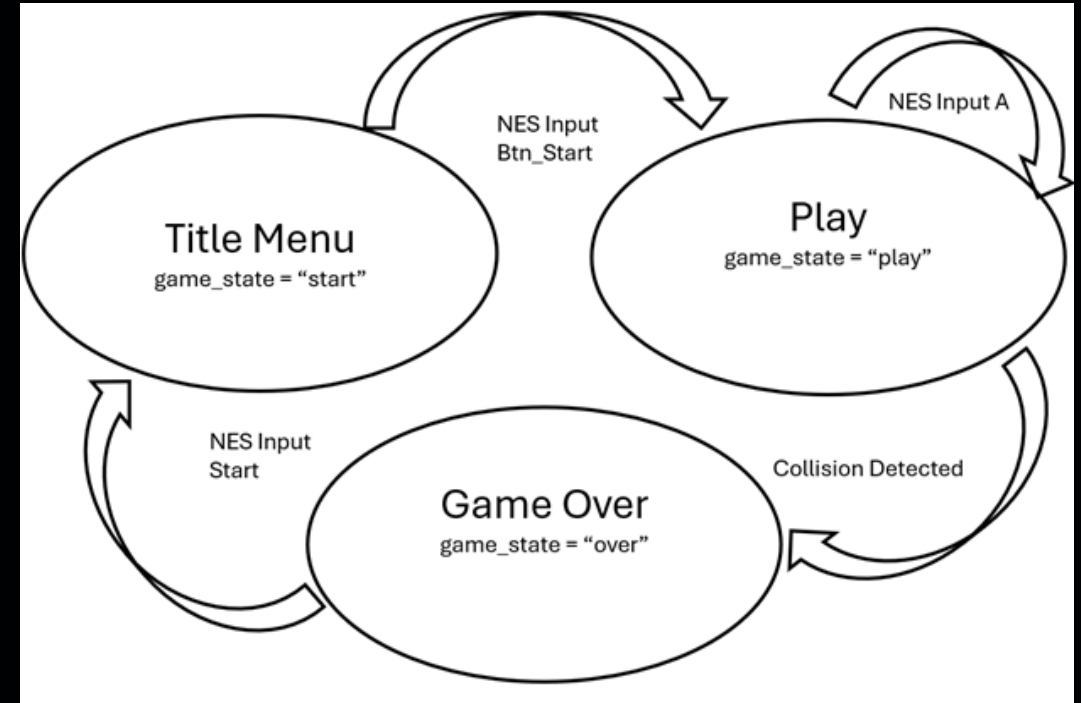
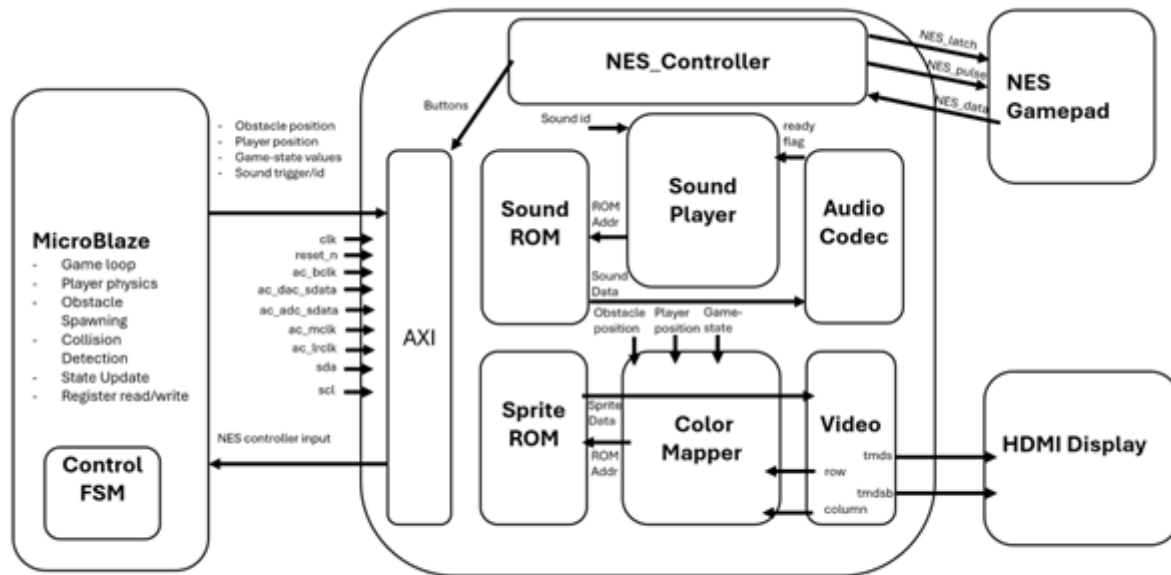
LIFE WITHOUT JETPACK JOYRIDE



FEATURES TO IMPLEMENT

- Milestone I
 - Render a static scene with all sprites in fixed positions:
 - Backdrop
 - Barry
 - Lasers
 - missiles.
 - Used to confirm sprite logic and VGA/video functionality.
- Milestone II
 - Implement game state transitions and the scrolling effect
 - Lasers move across the screen at the target speed, laser and missile spawning is handled correctly
 - collision logic plus score updates work as expected.





TECHNICAL APPROACH

4 PRIMARY COMPONENTS:

- COLOR MAPPER
- SOUND PLAYER
- NES CONTROLLER
- C CODE:
 - COLLISION
 - PHYSICS



Used for sprite rendering, sound playback, and input from NES gamepad

AXI	Register	Field used	Meaning
0x2C	slv_reg11	[9:0]	barry_x
0x30	slv_reg12	[9:0]	barry_y
0x34	slv_reg13	[9:0]	zapper_x
0x38	slv_reg14	[9:0]	zapper_y
0x3C	slv_reg15	[9:0]	missile_x
0x40	slv_reg16	[9:0]	missile_y
0x44	slv_reg17	[7:0]	NES buttons (existing)
0x48	slv_reg18	[2:0]	game_state[1:0]: 00=START, 01=PLAYING, 10=GAME_OVER
0x54	slv_reg21	[3:0]	[0] = 1 to start, 0 to rearm; [3:1] = sound id



TESTING TIMELINE

1. First tested static sprite rendering
2. Added position registers for controlled sprite rendering
3. Added NES controller
4. Tested NES controller registers pipeline to move Barry
5. Added game state registers for menu title rendering
6. Added counter to color mapper
7. Added sound
8. Finalized game logic in C



QUESTIONS?